Immersion, Attention, and Collaboration in Spatial Computing: A Study on Work Performance with Apple Vision Pro

Carolin Wienrich and David Obremski

Fig. 1: The lowest level of immersion with the Apple Vision Pro while working on the collaborative task.

Abstract— Spatial computing is set to change the way we work. It will enable both focused work through a higher degree of immersion and collaborative work through enhanced integration of shared interaction spaces or interaction partners. With the Apple Vision Pro, the level of immersion can be adjusted seamlessly. So far, there have been no systematic studies on how this adjustability affects work performance when working alone or together. The present empirical study fills this research gap by varying the level of immersion across three stages (high, medium, low) while solving various tasks with the Apple Vision Pro. The results show that selective attention improves significantly with increasing immersion levels. In contrast, social presence decreases with increasing immersion. In general, participants performed better in the individual task than in the collaborative task. However, the degree of immersion did not influence the collaborative performance. In addition, we could not determine any adverse effects on depth perception or user experience after use. The present study provides initial contributions to the future of spatial computing in professional settings and highlights the importance of balancing immersion and social interaction in a world where digital and physical spaces seamlessly coexist.

Index Terms—Spatial Computing, Augmented Reality, Immersion, Collaborative Work, Selective Attention

1 Introduction

The Mixed Reality Continuum, introduced by Milgram et al. [18], categorizes technologies based on their level of virtuality, positioning fully immersive VR at one end and physical reality at the other. In between, various forms of augmented reality (AR) and augmented virtuality (AV) are defined. Recent extensions of this continuum focus more on the human experience, incorporating factors such as coherence, presence, and plausibility [24, 35]. These approaches delve deeper into the psychological effects of simultaneously experiencing both virtuality and reality on the user.

With new technological advancements, the seamless transition between VR and reality is becoming increasingly tangible. This is exemplified by optical and video see-through devices, which enable a switch between VR and AR modes on a single device [1]. The Apple Vision Pro, for instance, advertises its ability to adjust the level of immersion fluidly, positioning itself as a mixed reality (or spatial computing) device that can be used continuously in different work modes [10]. For cognitively demanding tasks that require individual focus, a fully immersive mode may be ideal as it minimizes distractions from the surrounding environment, which is particularly relevant in noisy settings like open-plan offices or co-working spaces [22]. For collaborative tasks, however, a more open mode that allows greater awareness of the physical environment may be more appropriate [8].

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While there are preliminary studies on see-through devices that explore foundational concepts such as coherence, plausibility, presence, and depth perception [16, 32–34], to our knowledge, no systematic investigation has examined how different immersion modes of the Apple Vision Pro impact both objective and subjective performance in various modes of work. This leads us to the following research questions:

1. Does the level of immersion influence task performance when working individually?
2. Does the level of immersion influence task performance when working collaboratively?
3. Does the usage of Apple Vision Pro cause negative effects after use?

To address these questions, we selected three levels of immersion with the Apple Vision Pro (0%, 50%, 100%) and empirically tested their effects on performance in a Stroop test [30] and a spot-the-difference task. The spot-the-difference task was performed both individually and collaboratively. Furthermore, depth perception was measured before and after using the Apple Vision Pro, and the interaction with the device was qualitatively assessed. The study was conducted in a laboratory designed to simulate a co-working space, complete with simulated street noise and pedestrians.

This paper contributes to the early exploration of how well the Apple Vision Pro supports various work tasks and modes, and how it may drive the promised transformation of the workplace through spatial computing [22].

2 Related Work

Advancements in Extended Reality (XR) technologies, encompassing VR, AR, and mixed reality (MR), have evolved significantly over the past decade [6]. Devices like the Oculus Quest series [19], Meta Quest Pro [17], and Meta's Ray-Ban Stories [21] have made immersive experiences more accessible to consumers. In addition, there is a growing interest in using XR for collaborative and productivity work environments [22].

2.1 The Further Development of Immersion and Presence in Mixed Reality

The Mixed Reality Continuum, first introduced by Milgram et al. [18], offers a foundational framework for understanding the spectrum between fully immersive virtual environments and physical reality. More recently, research has emphasized the importance of human experience within this continuum. Skarbez et al. [24] expanded on the concept by proposing that the quality of experience in mixed reality environments should not only be measured through presence but also through other dimensions such as coherence and plausibility. Wienrich et al. [35] deal with the transfer between the virtual and physical worlds, which refers to the mixing of virtual and physical stimuli to create a seamless experience. They combined the concept of continuum with the idea of *rest frames* introduced by Wirth et al. [36]. People have a sense of virtual presence if attention is allocated to immersive factors and the virtual environment is chosen as the primary reference frame. In addition to original ideas, real and virtual entities can possess a certain degree of referential power in mixed realities (MR) in the revised concept [35]. The entities can become referential cues determining, in turn, the selection of the referential frame and the emergence of presence in MR. Their findings emphasize the necessity of evaluating immersion not only through presence but also through these additional layers of subjective experience, providing a richer understanding of how immersion affects task performance and user satisfaction.

Slater's work has been particularly influential. He introduced the concepts of *place illusion* and *plausibility illusion*, arguing that for users to behave realistically in virtual environments, they must feel both physically present in the virtual space (place illusion) and believe that the events unfolding around them are actually happening (plausibility illusion) [25]. This dual focus on presence and plausibility has been critical in shaping our understanding of user engagement in immersive environments. Slater's studies found that users were more likely to exhibit realistic behavior in fully immersive environments when these two illusions were strong, contributing to the general sense of being 'in the moment'. In 2022, Slater updated these considerations after the increase in MR [26].

Further expanding on these concepts and closing the brackets to Skarbez, the *Congruence and Plausibility (CaP) model* explored the role of coherence and plausibility in virtual environments [13]. It highlighted that inconsistencies between virtual elements and the physical world can significantly disrupt user experiences. They found that higher levels of coherence, where virtual elements align closely with the user's real-world expectations and interactions, greatly enhance plausibility and user experience. This suggests that the seamless integration of both visual and behavioral coherence is crucial to maintain immersion and improve task performance in MR settings.

These works clarify that the experience in MR depends on how the virtual and physical worlds interact. The degree of immersion can be a useful manipulation for this.

2.2 The Impact of Immersion on Various User Experiences

Immersion refers to the objective characteristics of a virtual or mixed reality system that determine how thoroughly the system surrounds and engages the user's sensory inputs. It encompasses factors such as the field of view, display resolution, rendering fidelity, multi-sensory integration (e.g., visual, auditory, and haptic feedback), and the level of interactivity the system provides. High immersion systems maximize the sense of enclosure and reduce the perception of the physical environment by offering rich, high-fidelity stimuli across multiple sensory modalities [4, 28]. These systems are designed to create an all-encompassing experience, where users are primarily focused on the virtual environment due to the completeness and realism of the sensory input provided by the system [7].

2.2.1 Presence as Indicator for User Experience

Numerous studies have shown that the sense of presence increases with increasing immersion [14, 27]. Therefore, there is no doubt that presence is a good indicator or manipulation check to differentiate various levels of immersion. Assuming this, the degree of immersion will also influence other user states.

Lombard and Ditton [14] showed that increased presence leads to higher levels of user engagement and emotional involvement, which is essential for tasks that require deep concentration or emotional investment.

2.2.2 Performance as Indicator for User Experience

Cummings and Bailenson [7] conducted a systematic review focused on understanding the impact of immersive technology on presence and user experience. Their review included numerous studies investigating how the level of immersion in a virtual system (determined by factors such as display type, field of view, and sensory input) influences user experience and performance in different domains. They found better perceptual and cognitive performance with increasing immersion.

Similarly, other studies suggest that virtual environments can improve the focus by reducing distractions from the physical environment [15, 31]. Weibel et al. (2008) showed that users in higher immersion conditions reported a narrower focus and reduced perception of external stimuli, improving their flow experience and enjoyment [31]. Makrnsky et al. (2019) found that higher levels of immersion were correlated with fewer extraneous thoughts and distractions. The learners also reported that they were more focused on the virtual instructional content, suggesting potential performance gains.

Thus, researchers emphasized the potential of spatial computing devices in work-related tasks within noisy or distracting environments, such as open-plan offices or co-working spaces [22].

2.2.3 Social Aspects as Indicators for User Experience

However, increasing immersion may also introduce challenges related to communication cues and physical co-presence or social presence [9, 23]. Collaborative work in XR settings relies on the sense of social presence and the ability to effectively communicate and share virtual spaces [20].

Gutwin and Greenberg [9] analyzed the mechanisms of workspace awareness in collaborative environments. They highlighted how immersive systems might obscure important non-verbal communication cues, such as body language, eye contact, or subtle gestures, which are critical for maintaining smooth collaboration. In virtual environments, the lack of real-world physical presence can reduce the richness of communication and make it harder for collaborators to interpret each other's actions or intentions.

Schroeder [23] also explored the role of social presence in virtual environments, focusing on how immersive technologies can either enhance or diminish the feeling of 'being together' in a shared virtual space. While high immersion can create a stronger sense of presence by making users feel physically situated in the environment, it can also result in a sense of isolation if the system does not adequately simulate the social dynamics of real-world interaction. For example, even with advanced avatars or virtual representations, the limitations in conveying facial expressions or subtle movements can lead to misunderstandings or hinder effective communication in a group setting. These assumptions can be supported by a study by Steed et al. [29]. It has also shown that using avatars in immersive settings, while improving the sense of co-presence, can introduce challenges when avatars fail to replicate users' body language and facial expressions reliably. This misalignment between physical actions and their representation in the virtual world can impair team dynamics, making it harder for collaborators to build trust and understanding .

Oh et al. [20] conducted a systematic review on social presence in collaborative virtual environments and found that higher immersion often correlates with enhanced feelings of togetherness. However, this is not always accompanied by improved communication or collaboration. They identified that while immersive environments can foster a sense of shared space, they often struggle to replicate the nuances of face-to-face communication. For example, timing cues, emotional expressions, and even proxemics (using space in communication) can be difficult to reproduce accurately in highly immersive virtual environments, leading to potential breakdowns in collaborative workflows.

Moreover, Biocca et al. [2] developed a model of social presence that emphasizes the importance of mutual awareness and interaction fidelity in virtual spaces. Their work suggests that immersion alone is not enough to foster effective collaboration; the system must also support rich, contextually appropriate communication channels that allow users to convey subtle social signals. Without these, high immersion can actually detract from collaboration by creating a dissonance between the virtual environment and the social experience.

Therefore, while increased immersion enhances the feeling of physical presence in XR environments, it also introduces complexities in maintaining social presence and effective communication, which are vital for successful collaboration. These studies compare the aspects either in fully immersive VR, by toggling immersion cues on or off, or in classic AR interactions. The advantages and disadvantages in terms of social presence and collaboration have not yet been investigated in devices such as the Apple Vision Pro.

Furthermore, since the release of the device, many user reports have shared their experiences with the glasses. However, no scientific research exists on how the user experience is influenced. Some studies of other MR devices, for example, have shown impairments in depth perception during usage [16, 34]. However, it is not yet clear whether this is also impaired after the use.

2.3 Summary and Present Study

The reviewed literature reveals important insights into how immersion and presence influence user experience and performance in MR, including both individual and collaborative settings. Immersion has been shown to improve presence, with higher levels of immersion generally improving user focus, task performance, and emotional engagement, and partly also social presence [7, 14, 20, 27]. At the same time, there are challenges related to communication, social presence, and the nuances of non-verbal cues such as body language and facial expressions in immersive environments [9, 23, 29]. Despite these advancements, existing research primarily examines either fully immersive virtual reality (VR) or classic augmented reality (AR) systems, and there is a notable lack of studies on MR devices that allow for adjustable immersion levels, such as the recently released Apple Vision Pro.

A significant research gap lies in understanding how gradual modulation of immersion, ranging from 0% (full AR) to 100% (full VR), influences both individual task performance and collaborative work. Most prior studies have investigated immersion as a binary factor (high vs. low immersion) without considering the flexibility of newer MR devices that allow users to dynamically adjust the level of immersion based on the task at hand. In addition, little research examines performance on a task that is performed alone or together.

Furthermore, while previous research on MR devices suggests that depth perception may be impaired in some cases [16, 34], it is not clear whether Apple Vision Pro, with its advanced optics and user-adjustable immersion levels, mitigates or exacerbates these issues.

To address these gaps, we propose an experiment that systematically examines the performance impact of varying immersion levels (0%, 50%, and 100%) in individual and collaborative tasks using the Apple Vision Pro. The study evaluates **objective performance metrics** (i.e., task completion alone: selective attention measured by the Stroop test; task completion alone vs. together: work performance captured by correctly identified differences in a spot-the-difference task) and **subjective measures** (i.e., task completion together: sense of social presence measured by a questionnaire. In addition, we conducted two **manipulation checks** (environment perception and sense of presence at different levels of immersion), measured **depth perception** before and after usage, and assessed qualitative feedback about the **user experience**.

Based on the literature, we have the following expectations:

- **Manipulation Check**: The manipulation check is considered successful if the proportion of the perceived environment decreases with increasing immersion and the sense of presence increases.
- **H1**: With an increasing level of immersion, the performance in selective attention increases.
- **H2**: With an increasing level of immersion, the performance in the spot-the-difference task increases when performed alone.
- **H3**: With an increasing level of immersion, the performance in the spot-the-difference task decreases when performed together.
- **H4**: With an increasing level of immersion, the sense of social presence decreases.

In addition, we check exploratively whether depth perception changes through the use of the Apple Vision Pro and how users evaluated the user experience.

By conducting this pilot study, we aim to contribute to a deeper understanding of how mixed reality devices like the Apple Vision Pro can support dynamic work environments, where users regularly switch between individual focus and team collaboration, thereby addressing the current gaps in the literature on immersion and performance in MR environments.

3 Method

3.1 Study Design

The study was conducted using a 2x3 mixed design to evaluate the influence of different levels of immersion on selective attention and the quality of collaborative work using the Apple Vision Pro mixed-reality headset. The independent variables consist of **the level of immersion** (0%, 50%, 100%) as a between-subject factor and **the type of task** (individual vs. collaborative) as a within-subject factor. The dependent variables are the *selective attention*, measured using the Stroop test, and the *work performance*, assessed by the number of correctly identified differences in a spot-the-difference task. The

different levels of immersion can be seen in Figure 1 (0%), Figure 2 (50%) as well as the highest immersion (100%) shown in Figure 3.

3.2 Material and Apparatus

The study used two Apple Vision Pro headsets (running visionOS 1.0), each configured with different levels of immersion. The immersion levels were manipulated using the Digital Crown feature on the device, which adjusts the level of the blend of virtual and real-world elements. The tasks were performed using headsets and desktop computers for questionnaire administration and data recording. The experiment also involved using a projector to simulate public spaces, enhancing the realism of the task environment, whose projection can be seen in the background of the collaborative task in Figure 4 and the individual task in Figure 5.

Fig. 2: The medium level of immersion doing the collaborative task.

3.3 Participants

30 participants were recruited from a pool of southern german university students ($M_{age} = 22.10, SD_{age} = 1.77$). 20 of them indicated to be of female, 10 of male gender. Every participant except one reported German as their mother tongue. All participants could be assigned to the Caucasian ethnic group, which makes up the majority of the local students. Each session involved two participants working in separate rooms equipped with the necessary hardware and software to ensure the integrity of the experimental conditions. The room design was identical. In the collaborative task, participants worked in one room together (see procedure). The participants were invited to the sessions in such a way that they did not see each other before the experiment or during individual tasks. They registered separately for the experiment and were not recruited in dyads. Further, participants were previously screened for any contraindications to virtual reality usage, such as a history of motion sickness or visual impairments that might interfere with the experiment. According to German regulations, this study does not require the approval of an ethics committee, as no patients or vulnerable groups were examined and no special emotional states were triggered. The experiment was carried out according to the guidelines of the ethics committee of the Institute for Human Computer Media at the University of Würzburg. The corresponding templates for the consent forms were used, which are based on the Helsinki Declaration and the GDPR. All subjects gave their written consent, were informed about the procedure and the use and storage of data, and had the option to discontinue the experiment at any time without giving reasons. No one discontinued the experiment or reported negative effects directly after the experiment.

3.4 Procedure

Upon arrival, participants were welcomed and briefed about the study objectives. They were then asked to complete a set of initial questionnaires, including demographic questions and the Virtual Reality Sickness Questionnaire (VRSQ) [12], which consisted of various symptoms (e.g., fatigue, headache) that were rated on a four-point Likert

Fig. 3: The highest level of immersion using the Apple Vision Pro.

scale that ranged from 'no' (0) to 'severe' (3). Following this, the participants completed a depth perception task to establish a baseline measure. In this task, they were asked to estimate the distance from themselves to various markers on the floor. The participants then donned Apple Vision Pro headsets, which were individually calibrated for eye and hand tracking. The experiment consisted of two primary tasks: the Stroop test and a spot-the-difference task. The Stroop test [11] was used to measure selective attention by presenting participants with a list of color words (e.g., red, blue), which corresponded to the colour of the word itself (congruent) or not (incongruent). Participants were asked to name the colour of the word and then the reaction time and error rates between both conditions were compared. The *spot-the-difference task* [5] was used to assess both individual and collaborative performance by asking participants to identify the differences between two presented images. In the individual condition, participants performed the spot-the-difference task alone. In the collaborative condition, they worked together with another participant through the shared virtual environment of Apple Vision Pro. Five differences per picture could be found. In both task types, they processed three pictures. After completing these tasks, the participants repeated the depth perception task to determine any changes due to the experimental conditions. Finally, participants completed a second set of questionnaires, including a follow-up VRSQ [12] and the Networked Minds Social Presence Inventory (NMSPI) [3], which consists of various items regarding social presence that are rated on a 7-point Likert scale, reaching from 'strongly disagree' (0) to 'strongly agree' (7). The study concluded with a debriefing session in which participants were thanked and informed about the study's goals.

3.5 Data Analyses

The data was then compiled and analyzed, focusing on the relationship between immersion levels and both selective attention and collaborative work quality. All calculations were done using R-4.4.1. A series of one-way ANOVAs were conducted to examine the effects of immersion on selective attention and work performance. Additionally, planned contrasts and t-tests were calculated to further explore the differences between the different levels of immersion.

4 Results

4.1 Manipulation Check

Both manipulation checks were significant as expected. Participants reported seeing less of the real environment with increasing levels of immersion, $F(2, 15.40) = 3.94$, $p = .04$, $\eta_p^2 = 0.20$, and also experiencing a greater sense of presence, $F(2, 17.08) = 10.41$, $p = .001$, $\eta_p^2 = 0.39$ (Figure 6).

4.2 Impact of Immersion on Performance when Working Alone

As expected, the degree of immersion has a significant impact on the performance of the **Stroop test**, $F(2, 27) = 4.58$, $p = .02$, $\eta_p^2 =$

Fig. 4: The participants working together on the collaborative task with a projector displaying distraction in the background.

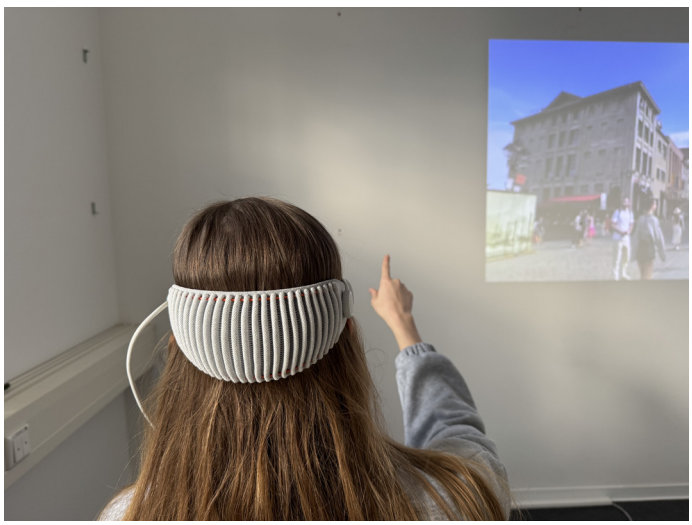

Fig. 5: A participant working on the individual task with a projector displaying distraction in the background.

0.25. Participants who work with a high degree of immersion ($M = 52.70$, $SD = 6.04$) perform significantly, $t(27) = 2.74$, $p = .01$, better than those with medium ($M = 48.10$, $SD = 4.46$) and low degrees of immersion ($M = 44.70$, $SD = 7.03$) (Figure 7).

4.3 Impact of Immersion on Performance and Perception when Working Together

As expected, the degree of immersion influences the **sense of social presence**, $F(2,15.18) = 6.46$, $p = .001$, $\eta_p^2 = 0.29$. The planned contrasts show that a high degree of immersion ($M = 4.84$, $SD = .43$) leads to less social presence compared to a medium degree ($M = 4.24$, $SD = 0.40$; $t(24) = 3.15$, $p = .004$) and a low degree ($M = 3.66$, $SD = 1.76$, $t(24) = 2.79$, $p = .010$). There is no difference between low and medium degrees $t(24) = 1.56$, $p = 1.32$ (Figure 8). Against the expectations, this did not correspond to the influence of immersion on **collaborative performance** ($F(2,27) = 2.37$, $p < .113$, $\eta_p^2 = 0.15$).

Furthermore, **fewer differences** were found during collaborative work ($M = 3.71$, $SD = 0.46$) than when working individually $M = 3.24$, $SD = 0.52$, $F(1,25) = 15.90$, $p < .001$, $\eta_p^2 = 0.39$). The interaction effect between the type of work and immersion is not significant, $F(2,25) = 2.74$, $p = .08$, $\eta_p^2 = 0.18$ (Figure 9). However, a planned contrast test showed that more errors were found in completing task alone compared to do it together under low and medium immersion conditions. In contrast, no differences occurred in the high immersion condition ($t(27) = 2.20$, $p = .037$).

The descriptive results are in line with these results and show

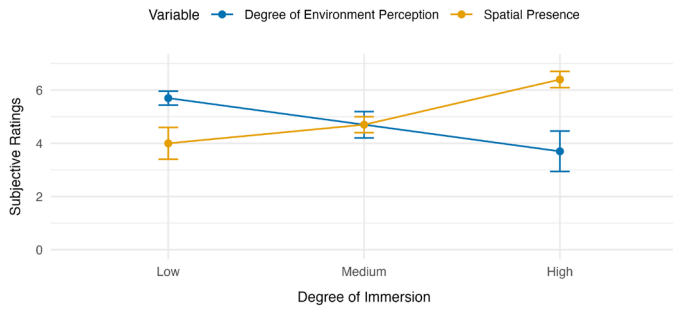

Fig. 6: The figure shows the results of the manipulation checks.

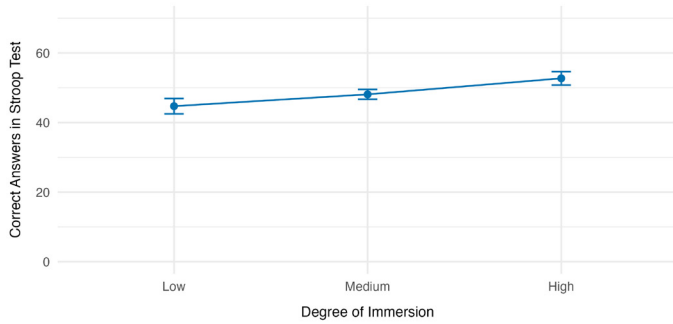

Fig. 7: The figure shows the results of the correct answers in the Stroop test.

different influences on immersion depending on the **type of work**. When working alone, more differences are found in the low immersion ($M_{ind} = 3.54$, $SD = 0.47$ versus $M_{tog} = 2.92$, $SD = 0.15$) and medium immersion condition ($M_{ind} = 4.00$, $SD = 0.35$ versus $M_{tog} = 3.27$, $SD = 0.72$) than working together. No descriptive differences were found in the high immersion condition ($M_{ind} = 3.57$, $SD = 0.36$ versus $M_{together} = 3.47$, $SD = 0.36$). In addition, there was no correlation between social presence and the number of differences found when the task was completed together ($r = -0.10$, $p = 0.569$.)

4.4 Usability and User Experience

The degree of immersion had no effect on depth perception, $F(2,27) = .46$, $p = .64$, $\eta_p^2 = 0.03$. Depth perception also did not change significantly before and after using the Apple Vision Pro, $F(1,27) = .24$, $p = .63$, $\eta_p^2 < 0.01$. There was also no interaction effect between immersion and measurement time, $F(2,27) = 1.03$, $p = .37$, $\eta_p^2 = 0.07$. (Figure 10).

Users rated overall use as positive and particularly mentioned fast response time, very simple and intuitive handling, clean, and efficient eye tracking. Only the weight of the glasses was negatively rated. Furthermore, almost all participants could imagine using the glasses in the future (26 out of 30). Five of them if they could afford it.

5 Discussion

The present study investigated the impact of varying immersion levels on both individual and collaborative work performance using the Apple Vision Pro. Our findings demonstrate a significant relationship between immersion and task performance, particularly in tasks requiring focused attention, while also highlighting the challenges that arise in maintaining social presence in collaborative settings. Furthermore, no side effects from use and a good user experience could be found.

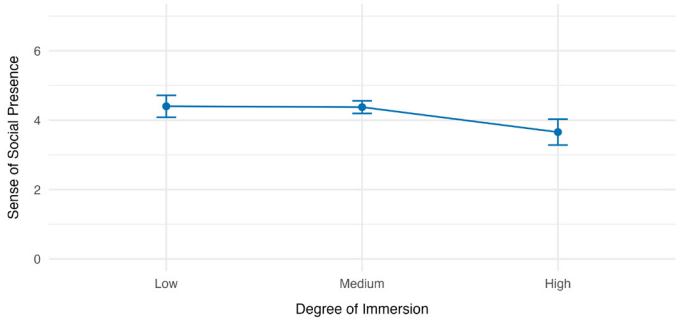

Fig. 8: The figure shows the results for the perceived sense of social presence.

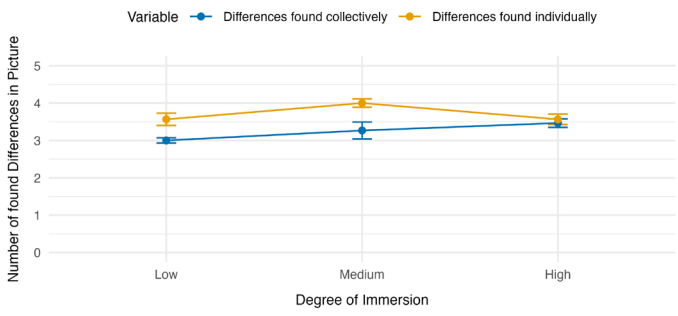

Fig. 9: The figure shows the results of the founded differences in the spot the differences task.

5.1 Immersion and Individual Task Performance

As hypothesized, the results revealed that higher levels of immersion led to improved selective attention in individual tasks. Participants in the high-immersion condition outperformed those in the medium- and low-immersion conditions on the Stroop test, supporting previous findings by Cummings and Bailenson [7] that increased immersion can improve cognitive performance by providing a distraction-free environment. This is also consistent with the work of others, who found that immersive virtual environments block out real-world stimuli, allowing better focus in noisy or distracting environments [15, 22, 31]. We simulated a co-working environment with slight distractions from pedestrians and street noise and thus showed that the results hold up under more realistic conditions. Presumably, a high degree of immersion is even more important when more distraction is added.

5.2 Immersion and Collaborative Task Performance

In contrast to the individual task, the results showed that increased immersion negatively impacted social presence during collaborative work. Participants in the high-immersion condition reported a significantly lower sense of social presence compared to those in the medium- and low-immersion conditions, supporting earlier findings by Gutwin and Greenberg [9] and Schroeder [23], which suggested that immersive systems can obscure important non-verbal communication cues, making it harder for users to interpret each other's actions and intentions.

Contrary to our expectations, the results showed that immersion has no impact on the performance in the collaborative task and that participants performed worse in the collaborative condition compared to when they worked individually. This is surprising, as previous literature generally suggests that collaborative work tends to result in better performance due to shared efforts and mutual support [20]. One possible explanation for this unexpected outcome could lie in the nature of the task used in this study. Although the task was modeled after the well-established collaborative spot-the-difference task [5], it was not an exact replication, which may have influenced the results. Another

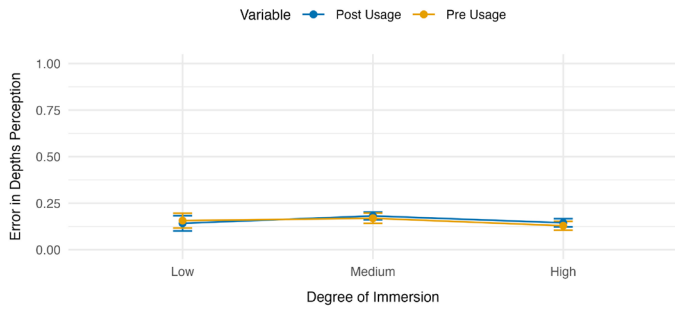

Fig. 10: The figure shows the deviation of depth perception

potential reason could be that the type of collaboration required by the task did not align well with the interaction capabilities of the Apple Vision Pro. For example, the task may not have fully leveraged the interactive features of the device, which are better suited for tasks that require more dynamic spatial interaction, such as collaborative problem solving in 3D environments or shared manipulation of virtual objects [2, 9]. Future studies should consider testing different types of collaborative tasks, to check whether our results are valid over a wider range of collaboration. Other studies have already found deficits in replicating the nuances of face-to-face communication, such as timing cues and proxemics [2, 20]. Thus, it would also be interesting to investigate communicative aspects further, e.g., with the EyeSight feature of the Apple Vision Pro. Furthermore, no diverse patterns of behavioral interaction during collaboration or a correlation between social presence and task performance were observed. These results also suggest that future studies should test collaborative tasks that encourage more interactive behavior. This could also be related to the fact that there were no clear first frames or referential cues for the collaboration here [35, 36]. In general, however, it is still unclear which aspects of MRs serve as referential cues and what coherent and plausible interaction can look like when the physical and virtual worlds intermingle [13, 24, 26]. The Apple Vision Pro provides an excellent examination device for this, so that the new theoretical developments in this area can be further investigated empirically [24, 35]. Establishing why and how the different modes and features improve work experience and performance and linking them with current theories in the field is also an important step for the future [13].

5.3 After-Effects

Our study did not find significant changes in depth perception before and after using the Apple Vision Pro, nor did depth perception vary significantly between different levels of immersion. This contradicts earlier research by Meng and Richard [16] and Westermeier et al. [34], who found that depth perception can be impaired in mixed reality (MR) environments. One difference in the method so far is that we have not measured the depth perception during use, but before and after use without the headsets. Thus, to what extent the device and its optical properties are better per se than the properties of the headsets used in the other studies remains to be researched. We can only say that no negative after-effects were observed.

5.4 Limitations and Future Work

Although this study provides valuable insights into the effects of immersion on work performance, it is not without limitations. The study was conducted in a controlled laboratory environment, which may not fully replicate the complexities of real-world work settings. By including street noise and pedestrians, we have already brought it close to a co-working environment, but it would be interesting to explore this even more realistically, such as open-plan offices or co-working spaces. For example, unexpected interruptions could also be investigated.

Furthermore, the sample size was relatively small and limited in diversity, which limits the generalizability of the findings. Future

studies should include larger and more diverse participant samples.

The collaborative task was based on the literature but perhaps did not fit perfectly in our setting. Future studies should test collaborative tasks that encourage more interactive behavior. Furthermore, interaction could be better captured through interviews or observations, thus complementing quantitative data. It could be checked beforehand that the task correlates positively with social presence, so that this proxy measure can actually provide indications of the effect of individual conditions on cooperation or other social states.

Moreover, future research should further explore the trade-offs between immersion and social presence in MR environments, particularly with devices such as the Apple Vision Pro, which allow users to seamlessly adjust their level of immersion.

Understanding how users can optimize their immersion levels based on task requirements could lead to more flexible and effective MR systems. An experimental design that allows users to choose this setting themselves and set different levels would be desirable in future studies.

In addition, further studies should examine the long-term use of such devices and whether repeated exposure to varying immersion levels affects depth perception, collaboration, and overall user satisfaction. Just because we have not found any side effects does not mean that there are none. There are many critical nonscientific user reviews on usability and user experience problems with the Apple Vision Pro.

Overall, the results should be seen as pilot results and further research should be conducted to generalize our findings. This also applies to the device. Currently, the Apple Vision Pro is the most prominent device of its kind. However, the findings should also be verified for other devices and thus generalized for spatial computing in this way.

6 Conclusion

In conclusion, this study contributes to the growing body of research on mixed reality or spatial computing systems by providing empirical evidence on how varying immersion levels impact individual and collaborative work performance. While increased immersion improves focus and selective attention in individual tasks, it also presents challenges for social presence in collaborative environments. The results underscore the need for a careful balance between immersion and interaction fidelity to ensure optimal performance in mixed reality work spaces. In general, however, it is still unclear which aspects of MR serve as referential cues and what coherent and plausible interaction can look like when the physical and virtual worlds intermingle [13, 24, 26]. Current theoretical work in this area has so far paid little attention to interactive social aspects and tends to describe the experience of individual users in MR.

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